AVE/CLAW HACKATHON 2025

Ave Talon

Autonomous AI signal agent for Solana. Scans, detects, researches, and alerts — all in real-time.

5 DIMENSIONS · 10 SIGNAL PATTERNS · 22 AI TOOLS

Built on Ave's on-chain data infrastructure
Observer-only · No private keys · Pure signal intelligence

01 What is Ave Talon?

An autonomous AI agent that monitors Solana tokens and tells you what's happening — before the crowd notices.

Ave Talon is a **fully autonomous signal detection agent** built on top of Ave's on-chain data API. It continuously scans graduated pump.fun tokens, analyzes them across 5 key dimensions, classifies them into 10 distinct signal patterns, and generates AI research reports with real Twitter data.

Think of it as having a 24/7 crypto analyst that never sleeps — it watches holder movements, whale behavior, volume spikes, and price action all at once, then tells you in plain English what it found.

How It Works

SCAN	→	DETECT	→	RESEARCH	→	ALERT
Fetch tokens from Ave API		5 dimensions 10 patterns		Twitter search AI narrative		Terminal feed Telegram bot

Every **3 minutes**, the orchestrator runs a full scan cycle. It pulls the latest graduated tokens from Ave, builds data snapshots (holders, klines, volume), runs them through the signal detector, and for any entry signals found — searches Twitter for real social data and generates an AI-powered research card.

> **Observer-only by design.** Ave Talon never holds private keys and never executes trades. It returns unsigned swap transactions for wallet signing (e.g., Phantom). This is a deliberate architectural choice — the agent observes and advises, the human decides.

Tech Stack

`Next.js 16` App Router, server components	`Ave API` On-chain data, holders, klines
`PostgreSQL` Prisma ORM, signal persistence	`Internal MCP` Real Twitter/X search
`AI Narrative` Sonnet-powered analysis	`Telegram Bot` Real-time alert notifications

02 The 5 Dimensions

Every token is analyzed across 5 independent dimensions. Each dimension measures a different aspect of what's happening on-chain.

1. Holder Growth

Are new people buying this token? Measures the percentage change in total holder count over a 5-minute window.

```
growth = ((current_holders - previous_holders) / previous_holders) × 100
```

*If a token had **1,000 holders** five minutes ago and now has **1,050**, that's a **+5%** growth rate. Anything above +2% starts getting interesting.*

2. Top 10 Concentration

What are the biggest wallets doing? Tracks the change in the percentage of supply held by the top 10 wallets.

```
delta = current_top10_ratio - previous_top10_ratio
```

*If the top 10 wallets held **45%** of supply and now hold **42%**, that's a **-3%** delta — whales are distributing (selling to smaller holders), which can be bullish.*

3. Real Money (Holders > $10)

Filters out dust wallets. Counts only holders with more than $10 worth of the token — these are real buyers, not airdrop bots.

```
delta = current_holders_above_10usd - previous_holders_above_10usd
```

*If **200** wallets held >$10 worth and now **230** do, that's **+30** real money accounts entering. This filters noise from dusting attacks.*

4. Volume vs Average

Is the token trading more than usual? Compares current volume to the 1-hour average.

```
ratio = current_volume / average_volume_1h
```

*A ratio of **3.5×** means the token is trading at 3.5 times its usual volume. Above 2× signals unusual activity; above 5× is a volume explosion.*

5. Price Change

The bottom line — is the price going up or down? Percentage change over the same 5-minute window.

```
change = ((current_price - previous_price) / previous_price) × 100
```

A price move from **$0.001** *to* **$0.0013** *is a* **+30%** *change. Combined with the other 4 dimensions, this tells us if the move is real or just noise.*

03 10 Signal Patterns

The 5 dimensions combine to form 10 distinct patterns. Each pattern represents a specific market situation with a recommended action.

#	PATTERN	ACTION	WHAT IT MEANS
1	**Strong Entry**	ENTRY	All 5 dimensions are bullish. Whales distributing (selling to retail), price rising, volume active, new real holders joining. The best signal.
11	**Early Entry**	ENTRY	Quiet accumulation phase. Holders growing steadily, moderate volume, whales not dumping yet. Pre-pump territory — early bird signal.
7	**Sneak Entry**	ENTRY	A whale is slowly entering the token while holders remain stable. Low volume, under the radar — potential breakout setup.
2	**Wait (No Volume)**	WAIT	Holders are increasing but volume is quiet. Could turn into something, but no conviction yet. Watch and wait.
3	**FOMO Pump**	CAUTION	Price rising fast with retail FOMO. Volume high but driven by hype, not smart money. Late entry risk — the pump may already be priced in.
4	**Shakeout**	WAIT	Holders dropping but whales are still holding. Weak hands exiting while smart money stays. Could be a dip-buying opportunity.
8	**Divergence**	WAIT	Holders growing but price dropping — contradictory signals. New buyers are entering at lower prices. Unclear direction, needs more data.
10	**Pump + Whale Load**	CAUTION	Strong price pump AND whales are loading more. Sounds bullish but risky — whales might be preparing to dump after pumping.
5	**Whale Exit**	EXIT	Top wallets selling, high volume, declining holders. Smart money is leaving. If you're holding, consider exiting.
6	**Full Dump**	DANGER	Everything is red. Whales dumping, holders fleeing, volume high from panic selling, price crashing. Stay away.

How Scoring Works

Each signal gets a **confidence score from 0 to 100** based on three components:

■ SCORING FORMULA

BASE SCORE	Depends on confidence level — HIGH: 75, MEDIUM: 55, LOW: 35, NONE: 15
ACTION BONUS	ENTRY: +10, EXIT: -10, DANGER: -20, CAUTION: -5, WAIT: 0

MAGNITUDE	Each dimension contributes 0–5 bonus points based on how strongly it triggers (max +25)

```
final_score = clamp(base + action_bonus + magnitude_bonus, 0, 100)
```

A **Strong Entry** with HIGH confidence and extreme dimensions can score **100**. A weak Wait signal with no volume might score **35**.

04 AI Tool System

Ave Talon's AI agent has access to 22 specialized tools across 4 layers, all powered by Ave's on-chain data API.

LAYER 1 — CORE DATA (8 TOOLS)

`ave_get_token_detail`
Full token metadata, pair info, market data

`ave_get_holders`
Current holder list with balances and percentages

`ave_get_holder_timeseries`
Historical holder count over time windows

`ave_get_kline`
OHLCV candlestick data (1m to 1d intervals)

`ave_get_dev_info`
Developer wallet analysis and activity

`ave_get_risk_score`
Token risk assessment (0-100 score)

`ave_compute_mcap`
Calculate market cap from supply and price

`solana_wallet_balances`
SOL and SPL token balances for any wallet

LAYER 2 — SIGNAL ENGINE (5 TOOLS)

`signal_detect`
Run the 5-dimension detector on any token

`signal_get_recent`
Retrieve recent signals from the database

`signal_explain`
AI explanation of why a signal was triggered

`signal_score_dimensions`
Breakdown of each dimension's contribution

`signal_compare`
Compare signals between two tokens

LAYER 3 — SCANNER (3 TOOLS)

`scanner_trending`
Fetch current trending tokens on Solana

`scanner_filter`
Filter tokens by market cap, volume, holder count

`scanner_refresh`
Force a fresh scan of graduated tokens

LAYER 4 — WATCHLIST & TRADE (6 TOOLS)

`watchlist_add`

`watchlist_remove`

Add token to personal watchlist	Remove token from watchlist
`watchlist_list` View all watched tokens	`jupiter_quote` Get swap quote from Jupiter aggregator
`trade_propose` Create unsigned swap tx for wallet signing	`trade_history` View past trade proposals and confirmations

> All data comes from **Ave's on-chain data infrastructure** — holder timeseries, kline OHLCV data, token metadata, and risk scoring. The agent never accesses private keys or executes transactions directly.

05 Architecture

How the autonomous loop is orchestrated — from scan to alert.

```
Orchestrator (singleton, boots with Next.js server)
│
├─ Scheduler        // 3-min interval, manages scan cycle timing
├─ Engine           // Core loop: fetch → snapshot → detect → research → alert
├─ Dedup            // Prevents duplicate signals (10-min TTL per token)
└─ Terminal         // Server-side log buffer, bridged to UI via polling

Engine pipeline per cycle:
│
├─ 1. Fetch graduated tokens from Ave API
├─ 2. Build snapshots (holders, klines, volume)
├─ 3. Run 5-dimension detector on each token
├─ 4. Classify into signal patterns (1-10)
├─ 5. Persist signals + snapshots to PostgreSQL
│
└─ For ENTRY signals (#1, #7, #11):
   ├─ 6. Search Twitter via internal MCP
   ├─ 7. Generate AI narrative (Sonnet)
   ├─ 8. Save research to DB (unique per token)
   └─ 9. Send Telegram alert + emit research card
```

Data Flow

The server-side orchestrator emits log lines to a ring buffer (500 lines max). The client polls this buffer every 5 seconds via `/api/agent/orchestrator`. Research cards, cycle summaries, and status updates all flow through this bridge. Research is also persisted to PostgreSQL, so cards survive server restarts.

06 Coming Soon

The roadmap for Ave Talon's evolution from observer to autonomous trader.

PHASE 2 **Autonomous Trading**

The agent will execute trades on its own — buying entry signals and selling exit signals through Jupiter. Still wallet-signed, but the agent decides when to trade, what size, and which tokens to target.

PHASE 3 **Self-Decision Making**

Advanced AI reasoning that weighs multiple signals, market context, portfolio exposure, and risk tolerance to make nuanced trading decisions. The agent will develop its own trading style based on configurable risk profiles.

PHASE 4 **Learning from Mistakes**

Outcome tracking is already built into the schema (1h, 4h, 24h price snapshots after each signal). The agent will use this historical performance data to tune its own thresholds, learn which patterns work best in different market conditions, and continuously improve its signal accuracy.

> Outcome tracking fields (`priceAt1h`, `priceAt4h`, `priceAt24h`, `outcomeScore`) are already in the database schema — the learning infrastructure is ready.

07 Summary

AVE TALON AT A GLANCE

What	Autonomous AI signal agent for Solana
Dimensions	5 (holder growth, top10, real money, volume, price)
Patterns	10 (3 ENTRY, 2 WAIT, 2 CAUTION, 1 EXIT, 1 DANGER, 1 WAIT)
Scan Cycle	Every 3 minutes, fully autonomous
AI Tools	22 tools across 4 layers
Data Source	Ave API (on-chain data infrastructure)
Twitter Research	Real tweets via internal MCP
Notifications	Telegram Bot API
Database	PostgreSQL via Prisma ORM
Framework	Next.js 16 (App Router)
Mode	Observer-only (no private keys)

Ave Talon demonstrates that autonomous AI agents can meaningfully process on-chain data to produce actionable trading intelligence. By combining Ave's data infrastructure with multi-dimensional signal detection, real social media research, and AI narrative synthesis, we've built an agent that doesn't just show you numbers — it tells you what they mean.

Ave Talon

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github.com/ShallIfy/ave-talon